

# PanX Platform Refinements

## Comprehensive Technical Development Scope & Verification Test Plan

Target System: PanX Web & Mobile Platforms  
(qsi.africa)

Document Version: v2.4 (Enterprise  
Baseline)

Theme Standard: Pan African Green  
(#008751)

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Design Source: Astro Designs (Figma  
Master)

Status: Approved for Implementation

## 1. Executive Summary & Architecture Context

This document defines the end-to-end development requirements and quality assurance acceptance criteria for the PanX platform overhaul. The scope spans navigational refactoring, global design system alignment, community feed enhancements (Threads model), creator and profile identity architectures, PanX Lab module modernizations, PanX TV and live streaming frameworks (TikTok paradigm), unified direct messaging (PanX Chats), and an enterprise multi-role SENSOL Task Card lifecycle state machine.

## 2. Detailed Development Scope

### 2.1 Global Navigation, Branding & Design System Standard

- **Global Search Renaming:** Re-label the primary search input placeholder across all platform headers from Search Feed to Search PanX.
- **Top Navigation Refactoring:** Update main navigation modules strictly to: PanX Feed, PanX Lab, PanX Concepts, and PanX TV.
- **PanX Mobility Transition:** Remove PanX Mobility from the public primary navigation bar and replace it with PanX Concepts. Preserve internal accessibility of PanX Mobility logic within the SENSOL Task Card architecture and administrative workflows.
- **Motion & Aesthetics:** Remove all non-functional icon and tab animations. Replace legacy icons with refined SVGs from the Astro Designs library.
- **Header Depth Consistency:** Enforce a fixed, uniform header strip height across all sub-applications (Feed, Lab, TV, Profiles, and Chats).
- **Pan African Green Palette:** Standardize all interactive accents, primary buttons, active indicators, and active badges to official Pan African Green (#008751).
- **Typography & Capitalization:** Deprecate ALL CAPS on buttons and tab selectors; enforce Title Case across all UI microcopy.

### 2.2 PanX Tools & SENSOL Module

- **Official SENSOL Nomenclature:** Rename Smart Infrastructure to SENSOL (Design. Plan. Execute.). Note the strict requirement of full stops after each word instead of commas.
- **Boundaries Architecture:** Refactor the PanX Tools layout to match the approved boundaries container layout (excluding PanX Mobility from public tools).
- **Visual Alignment:** Align the top margin/padding of the PanX Tools segment to match the vertical whitespace between the "Pan African Engineers" heading and its content. Synchronize typography styling and color between "Pan African Engineers" and "PanX Tools" headers.

## 2.3 PanX Feed & Social Architecture

- **Feed Filtering Engine:** Implement refined filter tabs: For PanX, For You, Following, and Friends. Deprecate non-functional filters.
- **Default Landing Experience:** Users default to For PanX, which aggregates content exclusively from Pan African Engineers and Sovereign Minds accounts. For You delivers blended social and platform-wide ecosystem content.
- **Filter Layout:** Apply left horizontal offset to leave sufficient breathing room before the first filter button.
- **Feed Spacing:** Reduce the vertical gap between the "What's on your mind" composer and the filter strip by 50%.
- **Threads-Style Post Display:** Refactor multi-image posts to support horizontal swipe/carousel interactions without image cropping or forced zooming. Implement clean, hierarchical comment tree architecture.
- **Media Publishing:** Enable multi-file simultaneous media uploads for Pan African Engineers profiles and grant profile owners full publishing privileges from their profile dashboards.

## 2.4 Profiles Architecture (Pan African Engineers & Sovereign Minds)

- **Verification Badges:** Deploy verified status badges for Pan African Engineers entities (Ventures and Sovereign Minds).
- **Social Graph:** Implement follow/unfollow functionality with dynamic real-time follow counters in profile headers.
- **Media Viewers:** Enable tap-to-expand lightbox view for all profile avatars.
- **Horizontal Sub-Navigation:** Introduce horizontal profile tabs: Posts, Project Ledger, and Sovereign Insights.
- **Role-Based Insight Publishing:** Restrict authoring within Sovereign Insights exclusively to Sovereign Minds profile types; enable content management on Project Ledger for engineering profiles.
- **PanX Concept Profile:** Re-skin PanX Concept profile views to conform seamlessly with the standard PanX profile design language.

## 2.5 PanX Lab (Learning & Studio Management)

- **Design Uniformity:** Match PanX Lab filter bar architecture to the refined PanX Feed design.
- **Taxonomy Filters:** Deploy category filters: Packages, For You, SENSOL, Digital Skills, Physics, etc.
- **Switchable "My Studio" Filter:** Integrate an interactive sub-state toggle directly inside the Packages button to switch between package catalog and creator studio (My Studio).
- **Package Card Architecture:** Re-engineer package cards with creator profile avatar, full name, tagline, and horizontally scrollable module listings.
- **Streamlined Enrollment:** Replace bulky "ENROLL MODULE" buttons with compact "Enroll" / "Get" buttons.
- **Curated Lab Feed:** Configure the Lab For You feed to aggregate user-relevant educational content (subscribed lectures, live sessions, course updates).
- **Healing & Wisdom Section:** Rename to Healing & Wisdom with tagline Cultivate the mind that builds. Incorporate updated Astro Design brand logo.
- **Suggestion Filtering:** Implement explicit filter tabs separating Suggestions from Packages rather than stacking them vertically.

## 2.6 PanX TV & Live Broadcast Engine

- **Header Standardization:** Match header height and styling to PanX Feed, enforcing solid monochrome text for "PanX TV" (no dual colors).
- **Control Bar Optimization:** Compact the Go Live button and nest Peer Session as an integrated sub-action to accommodate slim header heights.
- **Navigation Tabs:** Standardize filter tabs to: Live, For You, Channels, and My Studio. Refactor the For You interface using Apple TV-style horizontal carousel shelves.
- **Live Broadcast UI (TikTok Framework):**
  - Responsive multi-participant video grid layout with layout switcher (split screen, primary host, 4-grid).
  - Display live participant usernames directly over video tiles and within the chat feed.
  - Separate top broadcast controls from bottom chat inputs to eliminate viewport crowding.
  - Rename "LIVE INTELLIGENCE LOG" to PanX Live Chat. Relocate stream exit button from chat box to top status header (replacing 'active' badge).
  - Ensure opening chat on mobile does not occlude the primary video canvas.

## 2.7 PanX Chats (Direct Messaging & Support Stream)

- **Branding:** Rename system messaging from Inboxes to PanX Chats.
- **Segmented Inbox:** Introduce sub-filter toggles directly below search: Inbox and Requests.
- **Metadata & Timestamps:** Render sender profile pictures and compact timestamps (e.g. 14:20 or 12 Oct).
- **Thread Aggregation:** Consolidate automated system, SENSOL, and customer support notifications into a single persistent conversation thread per domain rather than creating redundant parallel chats.
- **Mobile Navigation Flow:** Configure mobile message navigation to load the full-screen conversations list directly instead of auto-opening previous active chats or sidebar drawers.

## 2.8 Unified SENSOL Task Card Lifecycle & State Machine

- **Component Redesign:** Rebuild Task Cards using Astro Design Figma specifications, unifying site visit requests (PanX Mobility) and engineering workflows (SENSOL).
- **Header Specification:** Rename card header from PanX Office @panx.engineering to Task Card @sustainable.engineering.solutions.
- **Engagement Model:** Remove irrelevant "Likes" counter; replace with task urgency, SLA timers, and progress status tags.

**Task Card Lifecycle State Machine:** Task cards are dynamically rendered across 5 distinct actor states based on user role and lifecycle progression.

### 1. Client / User

**Submit**

Generated via PanX AI chat. Client reviews, edits metadata, attaches files, and submits.

### 2. Client / User

**Track Task**

Post-dispatch state. Displays live telemetry, stage milestone, and engineer details.

### 3. Super Admin

**Assign Task**

Admin reviews scope, verifies feasibility, and delegates task to qualified professionals.

### 4. Professional

**Claim Task**

Assigned engineer receives task, reviews technical requirements, and claims ownership.

### 5. Professional

**Submit**

Execution complete. Engineer uploads deliverables, reports, and signs off milestone.

### 3. Verification & Quality Assurance Plan

The following test matrix defines the acceptance test cases required for QA sign-off across Web and Mobile viewports.

| Test ID    | Module / Feature                               | Execution Steps                                                                                                | Expected Acceptance Result                                                                                                                       |
|------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| TC-NAV-01  | <b>NAVIGATION</b><br>Global Tabs & Placeholder | 1. Load homepage.<br>2. Inspect search bar placeholder.<br>3. Inspect primary nav items.                       | Search placeholder reads "Search PanX". Nav tabs read: PanX Feed, PanX Lab, PanX Concepts, PanX TV. PanX Mobility is removed from public header. |
| TC-NAV-02  | <b>NAVIGATION</b> UI & Performance             | 1. Hover over tabs & click items.<br>2. Inspect icon assets.<br>3. Compare header height across routes.        | Animations disabled. Astro Design SVG icons render crisply. Header height is consistent across Feed, Lab, TV, and Chats.                         |
| TC-SYS-01  | <b>SYSTEM</b><br>SENSOL & Spacing              | 1. Navigate to PanX Tools.<br>2. Check SENSOL label punctuation.<br>3. Inspect top margin.                     | Label strictly formatted as "SENSOL (Design. Plan. Execute.)" with full stops. Whitespace matches Pan African Engineers segment.                 |
| TC-SYS-02  | <b>SYSTEM</b> Color & Typography               | 1. Inspect CSS variables.<br>2. Inspect filter buttons and CTAs.                                               | Primary accent is Pan African Green (#008751). All buttons/tabs use Title Case capitalization without ALL CAPS.                                  |
| TC-FEED-01 | <b>FEED</b> Default Landing & Filters          | 1. Open /feed.<br>2. Inspect active filter tab.<br>3. Switch between tabs.                                     | Default active tab is "For PanX", displaying Pan African Engineers & Sovereign Minds posts. "For You", "Following", "Friends" filter correctly.  |
| TC-FEED-02 | <b>FEED</b> Media Carousel (Threads)           | 1. Upload post with 4+ images.<br>2. Swipe through images on mobile and desktop.<br>3. Open comment thread.    | Smooth horizontal carousel swipe. Images retain native aspect ratio without forced zoom/cropping. Hierarchical comment tree renders cleanly.     |
| TC-FEED-03 | <b>FEED</b> Composer Spacing                   | 1. Measure whitespace between composer and filter tabs.                                                        | Vertical margin reduced by 50%, establishing balanced visual hierarchy.                                                                          |
| TC-PROF-01 | <b>PROFILES</b><br>Badges & Lightbox           | 1. View Pan African Engineer & Sovereign Minds profiles.<br>2. Click avatar.<br>3. Follow user.                | Verification badge displays. Avatar opens in full-screen lightbox. Follow count increments instantly without page reload.                        |
| TC-PROF-02 | <b>PROFILES</b> Tabs & RBAC Access             | 1. Check profile sub-navigation.<br>2. Attempt to publish Sovereign Insight as regular user vs Sovereign Mind. | Tabs: Posts, Project Ledger, Sovereign Insights. Sovereign Insights creation is restricted strictly to verified Sovereign Minds accounts.        |
| TC-LAB-01  | <b>LAB</b> Studio Switcher & Filters           | 1. Open PanX Lab.<br>2. Click "Packages" button switch toggle.<br>3. Inspect filter styling.                   | Filter styling matches PanX Feed. Clicking the internal Packages toggle seamlessly switches view to "My Studio".                                 |
| TC-LAB-02  | <b>LAB</b> Package Cards & CTA                 | 1. View course package card.<br>2. Scroll module list.<br>3. Click "Enroll".                                   | Card displays creator avatar, name, tagline, and horizontal module scroll. Compact "Enroll" CTA triggers enrollment modal.                       |
| TC-LAB-03  | <b>LAB</b> Healing & Wisdom                    | 1. Open Healing section.<br>2. Inspect title, tagline, logo.                                                   | Section titled "Healing & Wisdom", tagline reads "Cultivate the mind that builds", Astro Design logo rendered correctly.                         |
| TC-TV-01   | <b>PANX TV</b> Header & Controls               | 1. Navigate to PanX TV.<br>2. Inspect logotype and Go Live button.                                             | PanX TV text is solid monochrome. Go Live button is compact with integrated Peer Session dropdown option.                                        |
| TC-TV-02   | <b>PANX TV</b> Live Grid & Chat Overlay        | 1. Join live session with 3 participants.<br>2. Toggle grid layout.<br>3. Open chat on mobile.                 | Participant names display on video tiles. Layout adjusts smoothly. PanX Live Chat overlays cleanly without obscuring video stream.               |

| Test ID    | Module / Feature                             | Execution Steps                                                                                                                                                          | Expected Acceptance Result                                                                                                               |
|------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| TC-CHAT-01 | <b>PANX CHATS</b><br>Inbox & Filters         | <ol style="list-style-type: none"> <li>1. Navigate to PanX Chats.</li> <li>2. Toggle between Inbox &amp; Requests.</li> <li>3. Trigger support notifications.</li> </ol> | Header reads "PanX Chats". Filters work as expected. Support and SENSOL alerts aggregate into single unified conversation threads.       |
| TC-CHAT-02 | <b>PANX CHATS</b><br>Mobile Navigation       | <ol style="list-style-type: none"> <li>1. Set viewport to mobile.</li> <li>2. Click messages icon in bottom bar.</li> </ol>                                              | User lands directly on the full-screen conversations list view rather than an open chat or drawer.                                       |
| TC-TASK-01 | <b>TASK CARD</b><br>Client Creation & Submit | <ol style="list-style-type: none"> <li>1. Complete PanX AI prompt for site visit / SENSOL.</li> <li>2. Verify Task Card generation.</li> <li>3. Submit task.</li> </ol>  | Card header: "Task Card @sustainable.engineering.solutions". Action button is "Submit". Upon submission, button changes to "Track Task". |
| TC-TASK-02 | <b>TASK CARD</b><br>Admin Assignment         | <ol style="list-style-type: none"> <li>1. Log in as Super Admin.</li> <li>2. Open submitted Task Card.</li> <li>3. Assign to verified engineer.</li> </ol>               | Action button displays "Assign Task". Selecting professional dispatches task and updates status to Assigned.                             |
| TC-TASK-03 | <b>TASK CARD</b><br>Professional Execution   | <ol style="list-style-type: none"> <li>1. Log in as Assigned Professional.</li> <li>2. View card.</li> <li>3. Claim task.</li> <li>4. Submit deliverable.</li> </ol>     | Card displays "Claim Task". Once claimed, CTA changes to "Submit" for final sign-off and report attachment.                              |